# Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.47)

## Metric

$$ds^2 = \left(S^2h^4 + 4\,S^2h^3 + 6\,S^2h^2 + 4\,S^2h + S^2\right)\mathrm{d}r \otimes \mathrm{d}r + \left(-\frac{h^2 - 2\,h + 1}{h^2 + 2\,h + 1}\right)\mathrm{d}t \otimes \mathrm{d}t + \left(r^2S^2h^4 + 4\,r^2S^2h^3 + 6\,r^2S^2h^2 + 4\,r^2S^2h + r^2S^2\right)\mathrm{d}\theta \otimes \mathrm{d}\theta + \left(r^2S^2h^4 + 4\,r^2S^2h^3 + 6\,r^2S^2h^2 + 4\,r^2S^2h + r^2S^2\right)\sin\left(\theta\right)^2\mathrm{d}\phi \otimes \mathrm{d}\phi$$

## Coordinates

$(x^\mu) = (r, t, \theta, \phi)$

## Metric Functions

$$h = \frac{1}{\sqrt{\alpha r^2 + \beta}S(t)}$$

## Null Tetrad

*Covariant components*

$$l = \left(-\frac{1}{2}\sqrt{2}Sh^2 - \sqrt{2}Sh - \frac{1}{2}\sqrt{2}S\right)dr + \left(-\frac{\sqrt{2}h}{2\left(h+1\right)} + \frac{\sqrt{2}}{2\left(h+1\right)}\right)dt$$

$$n = \left(\frac{1}{2}\sqrt{2}Sh^2 + \sqrt{2}Sh + \frac{1}{2}\sqrt{2}S\right)dr + \left(-\frac{\sqrt{2}h}{2\left(h+1\right)} + \frac{\sqrt{2}}{2\left(h+1\right)}\right)dt$$

$$m = \left(\frac{1}{2}\sqrt{2}rSh^2 + \sqrt{2}rSh + \frac{1}{2}\sqrt{2}rS\right)d\theta + \left(\frac{1}{2}i\sqrt{2}rSh^2\sin\left(\theta\right) + i\sqrt{2}rSh\sin\left(\theta\right) + \frac{1}{2}i\sqrt{2}rS\sin\left(\theta\right)\right)d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}rSh^2 + \sqrt{2}rSh + \frac{1}{2}\sqrt{2}rS\right) d\theta + \left(-\frac{1}{2}i\sqrt{2}rSh^2\sin(\theta) - i\sqrt{2}rSh\sin(\theta) - \frac{1}{2}i\sqrt{2}rS\sin(\theta)\right) d\phi$$

*Contravariant components*

$$l = \left(-\frac{1}{\sqrt{2}Sh^2 + 2\sqrt{2}Sh + \sqrt{2}S}\right) dr + \left(\frac{h}{\sqrt{2}h - \sqrt{2}} + \frac{1}{\sqrt{2}h - \sqrt{2}}\right) dt$$

$$n = \left(\frac{1}{\sqrt{2}Sh^2 + 2\sqrt{2}Sh + \sqrt{2}S}\right) dr + \left(\frac{h}{\sqrt{2}h - \sqrt{2}} + \frac{1}{\sqrt{2}h - \sqrt{2}}\right) dt$$

$$m = \left(\frac{1}{\sqrt{2}rSh^2 + 2\sqrt{2}rSh + \sqrt{2}rS}\right) d\theta + \left(\frac{i}{\left(\sqrt{2}rSh^2 + 2\sqrt{2}rSh + \sqrt{2}rS\right)\sin(\theta)}\right) d\phi$$

$$\overline{m} = \left(\frac{1}{\sqrt{2}rSh^2 + 2\sqrt{2}rSh + \sqrt{2}rS}\right) d\theta + \left(-\frac{i}{\left(\sqrt{2}rSh^2 + 2\sqrt{2}rSh + \sqrt{2}rS\right)\sin(\theta)}\right) d\phi$$

## Directional Derivatives

$$D = -\frac{1}{\sqrt{2}Sh^2 + 2\sqrt{2}Sh + \sqrt{2}S}\,\partial_r + \frac{h}{\sqrt{2}h - \sqrt{2}} + \frac{1}{\sqrt{2}h - \sqrt{2}}\,\partial_t$$

$$\Delta = \frac{1}{\sqrt{2}Sh^2 + 2\sqrt{2}Sh + \sqrt{2}S}\,\partial_r + \frac{h}{\sqrt{2}h - \sqrt{2}} + \frac{1}{\sqrt{2}h - \sqrt{2}}\,\partial_t$$

$$\delta = \frac{1}{\sqrt{2}rSh^2 + 2\sqrt{2}rSh + \sqrt{2}rS}\,\partial_\theta + \frac{i}{\left(\sqrt{2}rSh^2 + 2\sqrt{2}rSh + \sqrt{2}rS\right)\sin(\theta)}\,\partial_\phi$$

$$\overline{\delta} = \frac{1}{\sqrt{2}rSh^2 + 2\sqrt{2}rSh + \sqrt{2}rS}\,\partial_\theta - \frac{i}{\left(\sqrt{2}rSh^2 + 2\sqrt{2}rSh + \sqrt{2}rS\right)\sin(\theta)}\,\partial_\phi$$

## Spin Coefficients

$$\rho = -\frac{\sqrt{2}rh^4\frac{\partial}{\partial t}S}{2\left(rSh^4+2\,rSh^3-2\,rSh-rS\right)} - \frac{\sqrt{2}rSh^3\frac{\partial}{\partial t}h}{rSh^4+2\,rSh^3-2\,rSh-rS} - \frac{2\,\sqrt{2}rh^3\frac{\partial}{\partial t}S}{rSh^4+2\,rSh^3-2\,rSh-rS} - \frac{3\,\sqrt{2}rSh^2\frac{\partial}{\partial t}h}{rSh^4+2\,rSh^3-2\,rSh-rS}$$
$$- \frac{3\,\sqrt{2}rh^2\frac{\partial}{\partial t}S}{rSh^4+2\,rSh^3-2\,rSh-rS} - \frac{3\,\sqrt{2}rSh\frac{\partial}{\partial t}h}{rSh^4+2\,rSh^3-2\,rSh-rS} - \frac{2\,\sqrt{2}rh\frac{\partial}{\partial t}S}{rSh^4+2\,rSh^3-2\,rSh-rS} - \frac{\sqrt{2}rS\frac{\partial}{\partial t}h}{rSh^4+2\,rSh^3-2\,rSh-rS} + \frac{\sqrt{2}rh\frac{\partial}{\partial r}h}{rSh^4+2\,rSh^3-2\,rSh-rS}$$
$$+ \frac{\sqrt{2}h^2}{2\left(rSh^4+2\,rSh^3-2\,rSh-rS\right)} - \frac{\sqrt{2}r\frac{\partial}{\partial t}S}{2\left(rSh^4+2\,rSh^3-2\,rSh-rS\right)} - \frac{\sqrt{2}r\frac{\partial}{\partial r}h}{rSh^4+2\,rSh^3-2\,rSh-rS} - \frac{\sqrt{2}}{2\left(rSh^4+2\,rSh^3-2\,rSh-rS\right)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}rh^4\frac{\partial}{\partial t}S}{2\left(rSh^4+2\,rSh^3-2\,rSh-rS\right)} + \frac{\sqrt{2}rSh^3\frac{\partial}{\partial t}h}{rSh^4+2\,rSh^3-2\,rSh-rS} + \frac{2\,\sqrt{2}rh^3\frac{\partial}{\partial t}S}{rSh^4+2\,rSh^3-2\,rSh-rS} + \frac{3\,\sqrt{2}rSh^2\frac{\partial}{\partial t}h}{rSh^4+2\,rSh^3-2\,rSh-rS} + \frac{3\,\sqrt{2}rh^2\frac{\partial}{\partial t}S}{rSh^4+2\,rSh^3-2\,rSh-rS}$$
$$+ \frac{3\,\sqrt{2}rSh\frac{\partial}{\partial t}h}{rSh^4+2\,rSh^3-2\,rSh-rS} + \frac{2\,\sqrt{2}rh\frac{\partial}{\partial t}S}{rSh^4+2\,rSh^3-2\,rSh-rS} + \frac{\sqrt{2}rS\frac{\partial}{\partial t}h}{rSh^4+2\,rSh^3-2\,rSh-rS} + \frac{\sqrt{2}rh\frac{\partial}{\partial r}h}{rSh^4+2\,rSh^3-2\,rSh-rS}$$
$$+ \frac{\sqrt{2}h^2}{2\left(rSh^4+2\,rSh^3-2\,rSh-rS\right)} + \frac{\sqrt{2}r\frac{\partial}{\partial t}S}{2\left(rSh^4+2\,rSh^3-2\,rSh-rS\right)} - \frac{\sqrt{2}r\frac{\partial}{\partial r}h}{rSh^4+2\,rSh^3-2\,rSh-rS} - \frac{\sqrt{2}}{2\left(rSh^4+2\,rSh^3-2\,rSh-rS\right)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\cos\left(\theta\right)}{2\left(\sqrt{2}rSh^2+2\,\sqrt{2}rSh+\sqrt{2}rS\right)\sin\left(\theta\right)}$$

$$\beta = \frac{\cos\left(\theta\right)}{2\left(\sqrt{2}rSh^2+2\,\sqrt{2}rSh+\sqrt{2}rS\right)\sin\left(\theta\right)}$$

$$\tau = 0$$

$\pi = 0$

$$\epsilon = \frac{\sqrt{2}h^4 \frac{\partial}{\partial t} S}{4\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} + \frac{\sqrt{2}Sh^3 \frac{\partial}{\partial t} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} + \frac{\sqrt{2}h^3 \frac{\partial}{\partial t} S}{Sh^4 + 2\,Sh^3 - 2\,Sh - S} + \frac{3\sqrt{2}Sh^2 \frac{\partial}{\partial t} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} + \frac{3\sqrt{2}h^2 \frac{\partial}{\partial t} S}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)}$$
$$+ \frac{3\sqrt{2}Sh \frac{\partial}{\partial t} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} + \frac{\sqrt{2}h \frac{\partial}{\partial t} S}{Sh^4 + 2\,Sh^3 - 2\,Sh - S} + \frac{\sqrt{2}S \frac{\partial}{\partial t} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} + \frac{\sqrt{2} \frac{\partial}{\partial t} S}{4\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} - \frac{\sqrt{2} \frac{\partial}{\partial r} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)}$$

$$\gamma = -\frac{\sqrt{2}h^4 \frac{\partial}{\partial t} S}{4\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} - \frac{\sqrt{2}Sh^3 \frac{\partial}{\partial t} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} - \frac{\sqrt{2}h^3 \frac{\partial}{\partial t} S}{Sh^4 + 2\,Sh^3 - 2\,Sh - S} - \frac{3\sqrt{2}Sh^2 \frac{\partial}{\partial t} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} - \frac{3\sqrt{2}h^2 \frac{\partial}{\partial t} S}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)}$$
$$- \frac{3\sqrt{2}Sh \frac{\partial}{\partial t} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} - \frac{\sqrt{2}h \frac{\partial}{\partial t} S}{Sh^4 + 2\,Sh^3 - 2\,Sh - S} - \frac{\sqrt{2}S \frac{\partial}{\partial t} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} - \frac{\sqrt{2} \frac{\partial}{\partial t} S}{4\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)} - \frac{\sqrt{2} \frac{\partial}{\partial r} h}{2\left(Sh^4 + 2\,Sh^3 - 2\,Sh - S\right)}$$

## Ricci Tensor Components

$$
\begin{aligned}
\Phi_{00} = {} & \frac{r h^8 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} - \frac{r S h^8 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{r S^2 h^6 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{r S^2 h^7 \frac{\partial^2}{(\partial t)^2} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{3 r h^7 \frac{\partial}{\partial t} S^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{3 r S h^7 \frac{\partial^2}{(\partial t)^2} S}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{r S h^6 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{6 r S^2 h^5 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& - \frac{5 r S^2 h^6 \frac{\partial^2}{(\partial t)^2} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{7 r h^6 \frac{\partial}{\partial t} S^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& - \frac{7 r S h^6 \frac{\partial^2}{(\partial t)^2} S}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{6 r S h^5 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{15 r S^2 h^4 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{9 r S^2 h^5 \frac{\partial^2}{(\partial t)^2} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{7 r h^5 \frac{\partial}{\partial t} S^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{7 r S h^5 \frac{\partial^2}{(\partial t)^2} S}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{15 r S h^4 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{20 r S^2 h^3 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& - \frac{5 r S^2 h^4 \frac{\partial^2}{(\partial t)^2} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{2 r S h^5 \frac{\partial^2}{\partial t \partial r} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& - \frac{2 r S h^4 \frac{\partial}{\partial t} h \frac{\partial}{\partial r} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{20 r S h^3 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{15 r S^2 h^2 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{5 r S^2 h^3 \frac{\partial^2}{(\partial t)^2} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{2 r S h^4 \frac{\partial^2}{\partial t \partial r} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{2 r h^4 \frac{\partial}{\partial t} S \frac{\partial}{\partial r} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& - \frac{4 r S h^3 \frac{\partial}{\partial t} h \frac{\partial}{\partial r} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{7 r h^3 \frac{\partial}{\partial t} S^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{7 r S h^3 \frac{\partial^2}{(\partial t)^2} S}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{15 r S h^2 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{6 r S^2 h \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{9 r S^2 h^2 \frac{\partial^2}{(\partial t)^2} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& - \frac{4 r S h^3 \frac{\partial^2}{\partial t \partial r} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{4 r h^3 \frac{\partial}{\partial t} S \frac{\partial}{\partial r} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& - \frac{7 r h^2 \frac{\partial}{\partial t} S^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{7 r S h^2 \frac{\partial^2}{(\partial t)^2} S}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} \\
& + \frac{6 r S h \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{5 r S^2 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2}
\end{aligned}
$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$
\begin{aligned}
\Phi_{11} = {} & \frac{r h^8 \frac{\partial}{\partial t} S^2}{4\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} - \frac{r S h^8 \frac{\partial^2}{(\partial t)^2} S}{4\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{r S^2 h^6 \frac{\partial}{\partial t} h^2}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& - \frac{r S^2 h^7 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{3\, r h^7 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& - \frac{3\, r S h^7 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{r S h^6 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{3\, r S^2 h^5 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2} \\
& - \frac{5\, r S^2 h^6 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{7\, r h^6 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& - \frac{7\, r S h^6 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{3\, r S h^5 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2} \\
& + \frac{15\, r S^2 h^4 \frac{\partial}{\partial t} h^2}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& - \frac{9\, r S^2 h^5 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{7\, r h^5 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& - \frac{7\, r S h^5 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{15\, r S h^4 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{10\, r S^2 h^3 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2} \\
& - \frac{5\, r S^2 h^4 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2\right)} \\
& + \frac{10\, r S h^3 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2\, r S^2 h^7 - 2\, r S^2 h^6 - 6\, r S^2 h^5 + 6\, r S^2 h^3 + 2\, r S^2 h^2 - 2\, r S^2 h - r S^2}
\end{aligned}
$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{rh^8 \frac{\partial}{\partial t} S^2}{2\left(rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2\right)} - \frac{rSh^8 \frac{\partial^2}{(\partial t)^2} S}{2\left(rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2\right)} \\
& + \frac{rS^2h^6 \frac{\partial}{\partial t} h^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} - \frac{rS^2h^7 \frac{\partial^2}{(\partial t)^2} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{3\,rh^7 \frac{\partial}{\partial t} S^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} - \frac{3\,rSh^7 \frac{\partial^2}{(\partial t)^2} S}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{rSh^6 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{6\,rS^2h^5 \frac{\partial}{\partial t} h^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& - \frac{5\,rS^2h^6 \frac{\partial^2}{(\partial t)^2} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{7\,rh^6 \frac{\partial}{\partial t} S^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& - \frac{7\,rSh^6 \frac{\partial^2}{(\partial t)^2} S}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{6\,rSh^5 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{15\,rS^2h^4 \frac{\partial}{\partial t} h^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} - \frac{9\,rS^2h^5 \frac{\partial^2}{(\partial t)^2} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{7\,rh^5 \frac{\partial}{\partial t} S^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} - \frac{7\,rSh^5 \frac{\partial^2}{(\partial t)^2} S}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{15\,rSh^4 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{20\,rS^2h^3 \frac{\partial}{\partial t} h^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& - \frac{5\,rS^2h^4 \frac{\partial^2}{(\partial t)^2} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} - \frac{2\,rSh^5 \frac{\partial^2}{\partial t \partial r} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{2\,rSh^4 \frac{\partial}{\partial t} h \frac{\partial}{\partial r} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{20\,rSh^3 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{15\,rS^2h^2 \frac{\partial}{\partial t} h^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{5\,rS^2h^3 \frac{\partial^2}{(\partial t)^2} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& - \frac{2\,rSh^4 \frac{\partial^2}{\partial t \partial r} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{2\,rh^4 \frac{\partial}{\partial t} S \frac{\partial}{\partial r} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{4\,rSh^3 \frac{\partial}{\partial t} h \frac{\partial}{\partial r} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} - \frac{7\,rh^3 \frac{\partial}{\partial t} S^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{7\,rSh^3 \frac{\partial^2}{(\partial t)^2} S}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{15\,rSh^2 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{6\,rS^2h \frac{\partial}{\partial t} h^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{9\,rS^2h^2 \frac{\partial^2}{(\partial t)^2} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{4\,rSh^3 \frac{\partial^2}{\partial t \partial r} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{4\,rh^3 \frac{\partial}{\partial t} S \frac{\partial}{\partial r} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& - \frac{7\,rh^2 \frac{\partial}{\partial t} S^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{7\,rSh^2 \frac{\partial^2}{(\partial t)^2} S}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} \\
& + \frac{6\,rSh \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2} + \frac{9\,rS^2 \frac{\partial}{\partial t} h^2}{rS^2h^8 + 2\,rS^2h^7 - 2\,rS^2h^6 - 6\,rS^2h^5 + 6\,rS^2h^3 + 2\,rS^2h^2 - 2\,rS^2h - rS^2}
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & \frac{r h^8 \frac{\partial}{\partial t} S^2}{4\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} + \frac{r S h^8 \frac{\partial^2}{(\partial t)^2} S}{4\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{2 r S h^7 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{3 r S^2 h^6 \frac{\partial}{\partial t} h^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{r S^2 h^7 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} + \frac{3 r h^7 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{3 r S h^7 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} + \frac{19 r S h^6 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{5 r S^2 h^5 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{5 r S^2 h^6 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{7 r h^6 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} + \frac{7 r S h^6 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{15 r S h^5 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{5 r S^2 h^4 \frac{\partial}{\partial t} h^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{9 r S^2 h^5 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} + \frac{7 r h^5 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& + \frac{7 r S h^5 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} + \frac{5 r S h^4 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{10 r S^2 h^3 \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} + \frac{5 r S^2 h^4 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{20 r S h^3 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{35 r S^2 h^2 \frac{\partial}{\partial t} h^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{5 r S^2 h^3 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} - \frac{7 r h^3 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{7 r S h^3 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} - \frac{51 r S h^2 \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{11 r S^2 h \frac{\partial}{\partial t} h^2}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{9 r S^2 h^2 \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{7 r h^2 \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} - \frac{7 r S h^2 \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{13 r S h \frac{\partial}{\partial t} S \frac{\partial}{\partial t} h}{r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2} - \frac{5 r S^2 \frac{\partial}{\partial t} h^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{5 r S^2 h \frac{\partial^2}{(\partial t)^2} h}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} - \frac{r h^3 \frac{\partial^2}{(\partial r)^2} h}{3\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} \\
& - \frac{3 r h \frac{\partial}{\partial t} S^2}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)} - \frac{3 r S h \frac{\partial^2}{(\partial t)^2} S}{2\left(r S^2 h^8 + 2 r S^2 h^7 - 2 r S^2 h^6 - 6 r S^2 h^5 + 6 r S^2 h^3 + 2 r S^2 h^2 - 2 r S^2 h - r S^2\right)}
\end{aligned}
$$

## Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$\Psi_2 = \frac{rh\frac{\partial}{\partial r}h^2}{rS^2h^7 + 5\,rS^2h^6 + 9\,rS^2h^5 + 5\,rS^2h^4 - 5\,rS^2h^3 - 9\,rS^2h^2 - 5\,rS^2h - rS^2} - \frac{rh^2\frac{\partial^2}{(\partial r)^2}h}{3\,(rS^2h^7 + 5\,rS^2h^6 + 9\,rS^2h^5 + 5\,rS^2h^4 - 5\,rS^2h^3 - 9\,rS^2h^2 - 5\,rS^2h - rS^2)}$$
$$+ \frac{h^2\frac{\partial}{\partial r}h}{3\,(rS^2h^7 + 5\,rS^2h^6 + 9\,rS^2h^5 + 5\,rS^2h^4 - 5\,rS^2h^3 - 9\,rS^2h^2 - 5\,rS^2h - rS^2)} - \frac{3\,r\frac{\partial}{\partial r}h^2}{rS^2h^7 + 5\,rS^2h^6 + 9\,rS^2h^5 + 5\,rS^2h^4 - 5\,rS^2h^3 - 9\,rS^2h^2 - 5\,rS^2h - rS^2}$$
$$+ \frac{rh\frac{\partial^2}{(\partial r)^2}h}{3\,(rS^2h^7 + 5\,rS^2h^6 + 9\,rS^2h^5 + 5\,rS^2h^4 - 5\,rS^2h^3 - 9\,rS^2h^2 - 5\,rS^2h - rS^2)} - \frac{h\frac{\partial}{\partial r}h}{3\,(rS^2h^7 + 5\,rS^2h^6 + 9\,rS^2h^5 + 5\,rS^2h^4 - 5\,rS^2h^3 - 9\,rS^2h^2 - 5\,rS^2h - rS^2)}$$
$$+ \frac{2\,r\frac{\partial^2}{(\partial r)^2}h}{3\,(rS^2h^7 + 5\,rS^2h^6 + 9\,rS^2h^5 + 5\,rS^2h^4 - 5\,rS^2h^3 - 9\,rS^2h^2 - 5\,rS^2h - rS^2)} - \frac{2\,\frac{\partial}{\partial r}h}{3\,(rS^2h^7 + 5\,rS^2h^6 + 9\,rS^2h^5 + 5\,rS^2h^4 - 5\,rS^2h^3 - 9\,rS^2h^2 - 5\,rS^2h - rS^2)}$$

$\Psi_3 = 0$

$\Psi_4 = 0$

## Newman-Penrose Equations

*Equations are vanished.*

## Bianchi Identities

*Equations are vanished.*

## Petrov Type

Type "$D$"